

Power Function example

Plotting power function

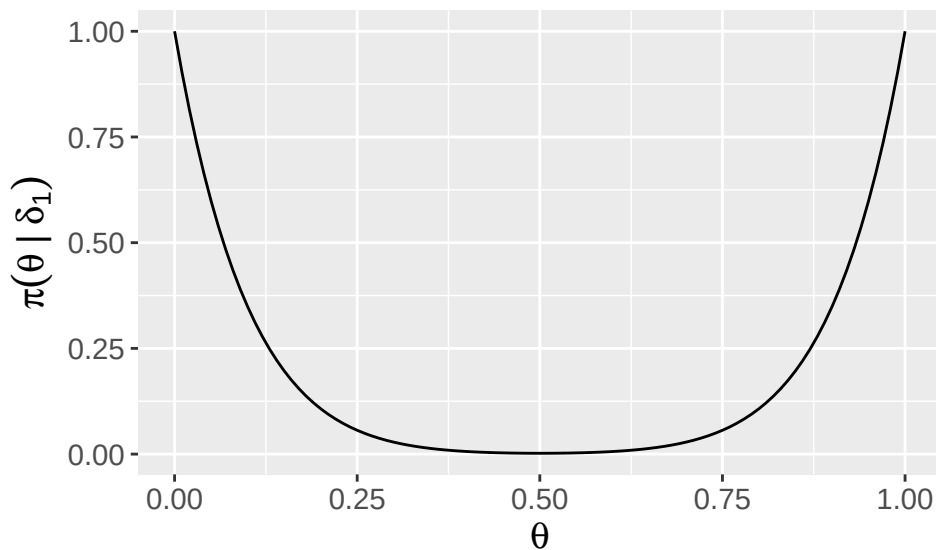
Recall the following statistical model and hypotheses:

$$X_1, \dots, X_{10} | \theta \sim \text{Bern}(\theta), \quad \theta \in [0, 1]$$

$$H_0 : \theta = \frac{1}{2} \quad \text{vs.} \quad H_1 : \theta \neq \frac{1}{2}$$

We have the following testing procedure δ_1 : $r(\mathbf{X}) = |\sum_{i=1}^{10} X_i - 5|$, with rejection region $R = \{r(\mathbf{X}) = 5\}$

So the power function is $\pi(\theta | \delta_1) = P(\sum_{i=1}^{10} X_i \in \{0, 10\} | \theta)$.

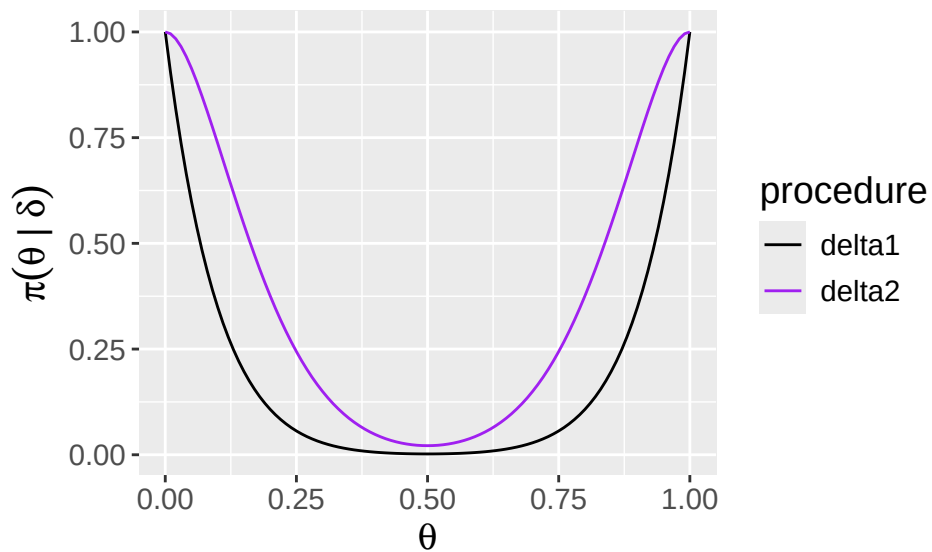


We can use the plot of this power function to talk about the probability of Type I and II error under δ_1 .

Comparing power

Consider the other testing procedure δ_2 : $r(\mathbf{X}) = |\sum_{i=1}^{10} X_i - 5|$ with rejection region $R = \{r(\mathbf{X}) \geq 4\}$.

So the power function is $\pi(\theta|\delta_2) = P(\sum_{i=1}^{10} X_i \in \{0, 1, 9, 10\}|\theta)$.



How do the two testing procedures δ_1 and δ_2 compare with respect to the probabilities of committing a Type I vs. Type II error?